Compatibility Report

Job Reference	**2026-027**
STBL	MT Diligent Warrior
SS	MT Sand

**All calculations are in Meters*

A. Hose Calculations

MT Sand

POINT A

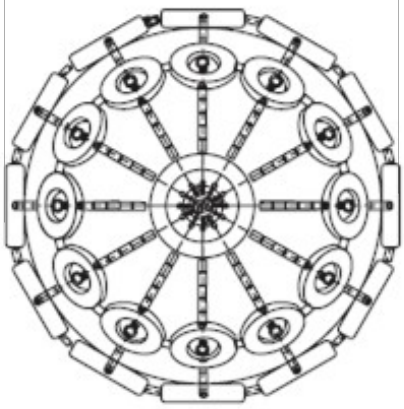
4.47

SS - Distance Manifold to Ship Rail

SS_MaxFreeboard	13.40
SS_MinFreeboard	7.30

POINT C

3.5 meters



MT Diligent Warrior

POINT B

4.40

STBL Distance Manifold to Ship Rail

STBL_Max_Freeboar	15.35
STBL_MinFreeboard	11.60

Hose Calculations

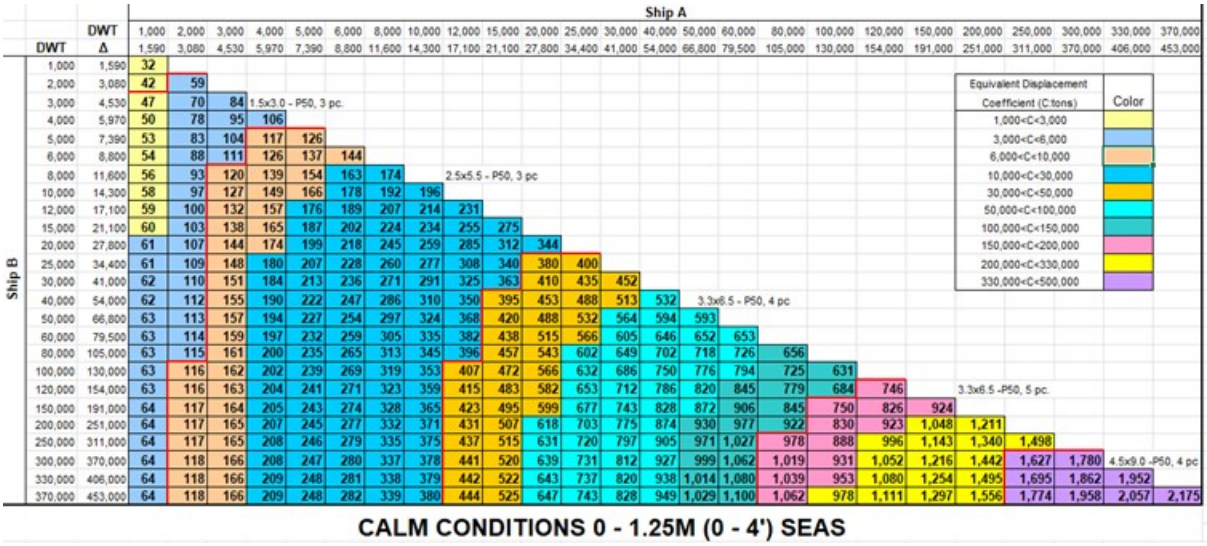
Max Freeboard	15.35
Min Freeboard	7.30
Freeboard Difference	8.05
Point A	4.47
Point B	4.40
Point C	3.5
Hose Length Required	20.4

Compatibility Report

B. Fender Calculations

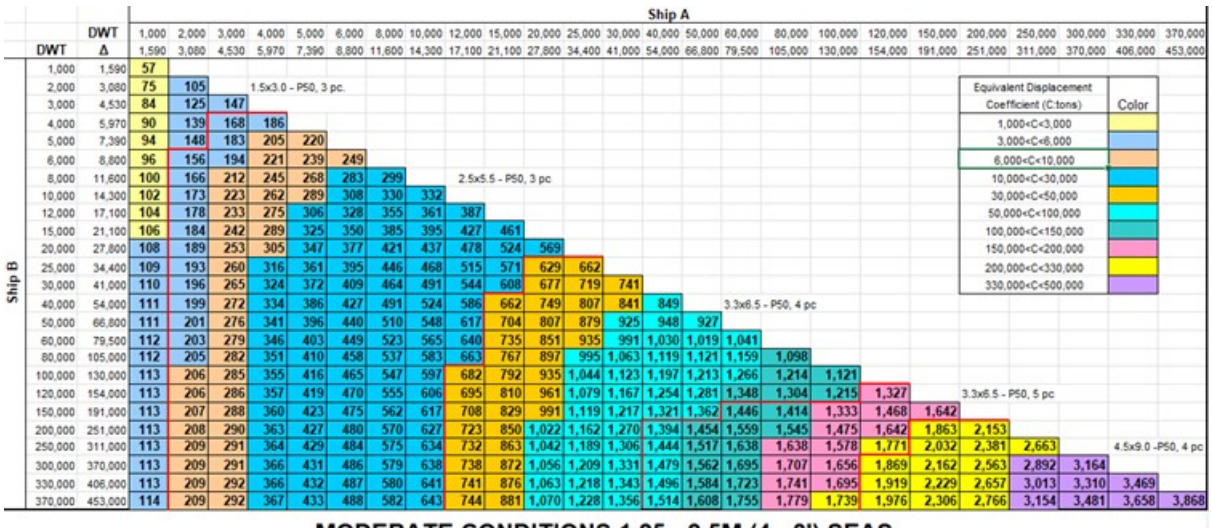
STBL_DWT	149992	MT	SS_DWT	115878	MT
STBL_Beam	48	Mtrs	SS_Beam	43.83	Mtrs
STBL_DISP	116824	MT	SS_Disp	63105	MT
STBL_Draft	11.5	Mtrs	SS_Draft	7.6	Mtrs

Equivalent Disp. Coefficient	81945.42	Combined virtual displacement	56969.90
STBL Mass Coefficient	1.48	Energy Coefficient Calm	1139.40
SS Mass Coefficient	1.35	Energy Coefficient Moderate	1780.31
Virtual Displacement STBL	172802.17	Energy Coefficient Rough	2563.65
Virtual Displacement SS	84989.46		



CALM CONDITIONS 0 - 1.25M (0 - 4') SEAS

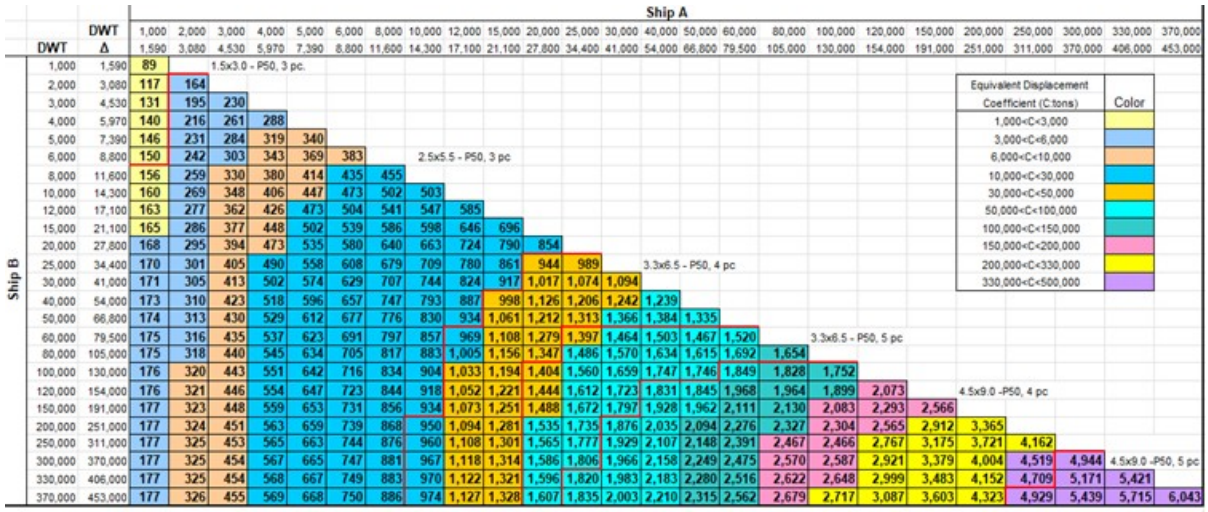
Fender required for Calm weather 4Nos x 3.3M x 6.5M



MODERATE CONDITIONS 1.25 - 2.5M (4 - 8') SEAS

Fender required for Moderate Sea 4Nos x 3.3M x 6.5M

Compatibility Report



Ship B DWT	Ship B Δ	Ship A DWT 1,000	Ship A Δ 1,590	2,000 / 3,080	3,000 / 4,530	4,000 / 5,970	5,000 / 7,390	6,000 / 8,800	8,000 / 11,600	10,000 / 14,300	12,000 / 17,100	15,000 / 21,100	20,000 / 27,800	25,000 / 34,400	30,000 / 41,000	40,000 / 54,000	50,000 / 66,800	60,000 / 79,500	80,000 / 105,000	100,000 / 130,000	120,000 / 154,000	150,000 / 191,000	200,000 / 251,000	250,000 / 311,000	300,000 / 370,000	330,000 / 406,000	370,000 / 453,000
1,000	1,590	89			1.5x3.0 - P50, 3 pc.																						
2,000	3,080	117	164																								
3,000	4,530	131	195	230																							
4,000	5,970	140	216	261	288																						
5,000	7,390	146	231	284	319	340																					
6,000	8,800	150	242	303	343	369	383		2.5x5.5 - P50, 3 pc																		
8,000	11,600	156	259	330	380	414	435	455																			
10,000	14,300	160	269	348	406	447	473	502	503																		
12,000	17,100	163	277	362	426	473	504	541	547	585																	
15,000	21,100	165	286	377	448	502	539	586	598	646	696																
20,000	27,800	168	295	394	473	535	580	640	663	724	790	854															
25,000	34,400	170	301	405	490	558	608	679	709	780	861	944	989		3.3x6.5 - P50, 4 pc												
30,000	41,000	171	305	413	502	574	629	707	744	824	917	1,017	1,074	1,094													
40,000	54,000	173	310	423	518	596	657	747	793	887	998	1,126	1,206	1,242	1,239												
50,000	66,800	174	313	430	529	612	677	776	830	934	1,061	1,212	1,313	1,366	1,384	1,335											
60,000	79,500	175	316	435	537	623	691	797	857	969	1,108	1,279	1,397	1,464	1,503	1,467	1,520		3.3x6.5 - P50, 5 pc								
80,000	105,000	175	318	440	545	634	705	817	883	1,005	1,156	1,347	1,486	1,570	1,634	1,615	1,692	1,654									
100,000	130,000	176	320	443	551	642	716	834	904	1,033	1,194	1,404	1,560	1,659	1,747	1,746	1,849	1,828	1,752								
120,000	154,000	176	321	446	554	647	723	844	918	1,052	1,221	1,444	1,612	1,723	1,831	1,845	1,968	1,964	1,899	2,073		4.5x9.0 -P50, 4 pc					
150,000	191,000	177	323	448	559	653	731	856	934	1,073	1,251	1,488	1,672	1,797	1,928	1,962	2,111	2,130	2,083	2,293	2,566						
200,000	251,000	177	324	451	563	659	739	868	950	1,094	1,281	1,535	1,735	1,876	2,035	2,094	2,276	2,327	2,304	2,565	2,912	3,365					
250,000	311,000	177	325	453	565	663	744	876	960	1,108	1,301	1,565	1,777	1,929	2,107	2,148	2,391	2,467	2,466	2,767	3,175	3,721	4,162				
300,000	370,000	177	325	454	567	665	747	881	967	1,118	1,314	1,586	1,806	1,966	2,158	2,249	2,475	2,570	2,587	2,921	3,379	4,004	4,519	4,944	4.5x9.0 -P50, 5 pc		
330,000	406,000	177	325	454	568	667	749	883	970	1,122	1,321	1,596	1,820	1,983	2,183	2,280	2,516	2,622	2,648	2,999	3,483	4,152	4,709	5,171	5,421		
370,000	453,000	177	326	455	569	668	750	886	974	1,127	1,328	1,607	1,835	2,003	2,210	2,315	2,562	2,679	2,717	3,087	3,603	4,323	4,929	5,439	5,715	6,043	

Equivalent Displacement Coefficient (C.tons)	Color
1,000<C<3,000	
3,000<C<6,000	
6,000<C<10,000	
10,000<C<30,000	
30,000<C<50,000	
50,000<C<100,000	
100,000<C<150,000	
150,000<C<200,000	
200,000<C<330,000	
330,000<C<500,000	

ROUGH CONDITIONS > 2.5M (> 8') SEAS

Fender required for Rough Weather	5Nos x 3.3M x 6.5M